FEDERAL SPECIFICATION

ILLUMINATORS; MICROSCOPE, 115 VOLT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers 115-volt microscope illuminators, complete.

1.2 Classification.

1.2.1 *Types.*—Microscope illuminators covered by this specification shall be of the following types, as specified in the invitation for bids:

Type I. —Substage, daylight, a.c.-d.c.
Type II. —Junior, darkfield illumination and general use, a.c.-d.c.
Type III.—Senior, darkfield illumination and general use, a.c.-d.c.
Type IV.—Adjustable, darkfield illumination and general use, a.c.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

2.1 Specifications.—The following Federal specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

W-R-151—Receptacles (Convenience-Outlets), Adapters, Attachment-Plug-Caps, Cord-Connector-Bodies, Current-Taps, Motor-Base-Plugs, and Plug-Bodies; 250-Volts.
W-S-890—Switches; Snap, Miscellaneous.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Other publication. — The following Underwriters' Laboratories, Inc., Standard, of the issue in effect on date of invitation for bids, forms a part of this specification:

Standard for Flexible Cord and Fixture Wire.

(Copies of Underwriters' Laboratories, Inc., standards may be obtained from the Underwriters' Laboratories, Inc, 207 E. Ohio Street, Chicago 11, Illinois; 161 Sixth Avenue, New York 13, New York, or 500 Sansome Street, San Francisco 11, California).

2.3 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Underwriters' Laboratories, Inc., Standard for Flexible Cord and Fixture Wire is applicable to types II, III, and IV illuminators, and is hereby made a part of this specification, except where in conflict with the requirements of this specification.

3.1.1.1 The bidder shall submit to the purchasing agency proof that the material or appliance he proposes to supply under this

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:51Z
Check the source to verify that this is the current version before use.

specification conforms to the standards of the Underwriters' Laboratories regarding fire and casualty hazards. The label or listing of the Underwriters' Laboratories will be accepted as conforming to this requirement.

3.1.1.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the purchasing agency that his material or appliance conforms to the published standard, including methods of test, of the Underwriters' Laboratories.

3.1.1.3 Compliance with the above preliminary requirements regarding fire and casualty hazards does not absolve the bidder from complete compliance with the other requirements of this specification in order to secure the acceptance of his material or appliance.

3.2 Type I.—Substage, daylight a.c.-d.c.

3.2.1 *Material.*

3.2.1.1 *Plastic.* — Plastic shall be impact-resisting phenol formaldehyde.

3.2.1.2 *Glass.*—Glass shall be daylight glass filter.

3.2.2 *Design.*—The illuminator shall consist of the following parts:

a. Plastic housing for bulb.
b. Daylight filter glass for housing.
c. Bulb.
d Insulated connecting wires.

3.2.2.1 *Housing construction.* — The housing shall be 3 to 3½ inches long, approximately 2 inches wide, and 2 inches high, and shall have a wall thickness of approximately ⅛ inch. The plastic housing design shall incorporate adequate facilities for ventilation, such as large circular baffle openings or a series of louvers, so that air is permitted to circulate freely through the lamp housing without loss of light. The interior shall be suitably molded to receive lamp screw shell and terminals

3.2.2.2 *Glass filter.*—Glass shall be approximately 45 mm by 45 mm. by 5 mm. Filter shall be ground on one side, having a color-temperature correcting capacity of 190 ± 8 mireds (see 4.3.1). Illuminator shall be so constructed as to hold the glass firmly in position.

3.2.2.3 *Lamp.*—The lamp shall be an incandescent type, 115 volt, 15 watt, S-11 clear bulb, 2¼ inches long overall, and shall have a candelabra screw base.

3.2.2.4 *Extension cord* — Extension cord shall be flexible, rubber-covered, two-conductor cord not less than 5 feet long with a suitable two-blade parallel, nonpolarized-type, attachment-plug cap. The attachment-plug cap shall conform to the applicable requirements of Federal Specification W-R-151.

3.2.2.5 *Finish* —Unless otherwise specified, the finish shall be smooth or mottled black.

3.3 Type II. — Junior, darkfield illumination and general use, a.c.-d.c.

3.3.1 *Material.*—Material shall be as hereinafter specified.

3.3.2 *Design.* — Illuminator shall be compact, standing approximately 6½ inches high, 6 inches front to back (exclusive of tilting knob), and 4½ inches wide. Overall front to back measurement, including tilting knob, shall be approximately 7½ inches. There shall be an adjustable heat-absorbing lens and bulb of the type hereinafter specified, contained in a removable plastic housing. Illuminator shall operate on 110 volts, both alternating and direct current.

3.3.3 *Base.*—Base shall be a metal casting measuring approximately 5¾ inches long and 4¼ inches wide. It shall be of sufficient weight and shall be designed to insure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

stability. The base shall be easily removed by screw arrangement to permit securing the lamp to an upright stand. The screw shall be wing type and shall have standard tripod threads. The base shall be equipped with four rubber knobs and shall be finished with not less than two coats of black, heat-resisting enamel.

3.3.4 *Removable lamp housing.* — Removable lamp housing shall be black plastic, of the mineral-filler, phenolic-formaldehyde type. The top shall have a series of parallel perforations to provide adequate ventilation for dissipation of heat. Immediately below the perforations there shall be a suitable removable metal baffle plate finished in the inner surface with a light absorbing black. Positions of lens shall be adjustable horizontally by sliding in and out in front of lamp. Adjustment shall be accomplished by a specially designed plastic handhold at the bottom of the front end of the lens tube. Handhold shall be designed so that there is a formed area into which the fingers can fit to make lens adjustment by sliding tube in and out. The housing shall be held to the housing base by means of two metal pins so designed as to fit into the housing base at the front corners and a plastic covered screw at the rear. The screw shall be so designed as to readily serve as a tilting knob.

3.3.5 *Housing base.*—Housing base shall be a metal casting of sufficient size and rigidity for the purpose intended, perforated to provide up-draft ventilation and finished with two coats of heat-resisting enamel. The upper front surface shall be so designed as to provide a suitable rest for the adjustable lens carrier. A single contact bayonet type socket shall be securely mounted on the base, so located as to permit maximum efficiency. A "U" shaped metal heat deflector, having double side walls with an air space between the walls, shall enclose the lamp bulb on three sides. The inner surface of the inside walls shall be finished with a light-absorbing black. An optical reflector shall be securely mounted on the rear wall of the heat deflector, so located as to provide maximum concentration of light rays. On the under side of the base housing there shall be a metal "U" bracket to fit the cast metal swivel block, keyed to the case base. The "U" bracket shall be attached to the swivel by means of a bolt provided with a winged nut, so designed as to provide an easy means of adjusting the tension for tilting the lamp up and down.

3.3.5.1 *Extension cord.*—An electric extension cord not less than 8 feet in length, measured from base, shall be provided. A handy snap switch of good quality and easy working design in accordance with Federal Specification W–S–890 shall be provided 8 to 10 inches from the lamp base. The cord between the lamp socket and the switch shall be asbestos-covered, heat-resisting, insulated electric cord. The remaining length of the cord shall be rubber insulated and shall have a standard male plug on the outer end, in accordance with Federal Specification W–R–151. The cord shall be securely attached to the under surface of the base to minimize wear on the insulation and strain on lamp-socket connections.

3.3.6 *Lens.*—Lens shall be so designed as to provide practically a parallel beam of light. The lens shall be pigmented for color correction to provide an approximate "white light." The lens shall be of the proper focal length and made of glass of heat absorbing quality designed to absorb approximately 75 percent of the infra-red rays.

3.3.7 *Lamp.*—Lamp shall be an incandescent type, 115-volt, 110-watt, G–16½ spotlight, 3 inches in overall length, with single contact bayonet candelabra base, average light-center length of 1⅜ inches, class C, CC–13 filament, with approximately 1600 lumens, and have an average life of 200 hours.

3.3.8 *Lens tube.*—The lens tube shall be of proper length and design for the intended

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

purpose. It shall be of metal and properly designed to telescope into the housing. The lens shall be held in position by spring wire or other suitable means. The front of the lens tube shall be provided with a molded plastic collar and a spring clip of such design as to adequately accommodate a 75 mm. square glass filter not exceeding 3.2 mm. in thickness, and a glass filter approximately 50 mm. square and 4.0 to 5.0 mm. in thickness. Unless otherwise specified, square filters shall not be furnished.

3.3.9 *Intensity of illumination.*—With the lens tube racked in, the lamp shall provide not less than 300 foot candles at a distance of 12 inches + ¼ − 0 inch from the front of the lens. With the lens tube racked full out, the lamp shall provide not less than 750 foot candles at a distance of 12 inches + ¼ − 0 inch from the front of the lens.

3.3.10 *Finish.*—Except as specified in 3.3.3 and 3.3.5, metal parts shall be nickel or chromium plated.

3.3.11 *Accessories.*—In addition to the bulb required as regular equipment, two extra bulbs shall be furnished as spare parts with each illuminator

3.4 Type III.—Senior, darkfield illumination and general use, a.c.-d.c.

3.4.1 *Material.*—Material shall be of high quality throughout and suitable in all respects for the purpose intended.

3.4.2 *Design.*—The lamp housing shall be constructed in a manner to allow for ventilating louvers over the major part of the surface. The housing shall be designed for ease in changing lamps. The reflector shall be mounted in such a manner that it can be easily adjusted in order that the image produced by the reflector can be focused in the plane of the filament and so positioned that when the illuminator is focused on a screen at 10 feet, the filament will appear to have eight strands, with alternate strands slightly less intense due to the reflection from the mirror. The exterior parts of the illuminator shall not exceed 72° C. and adjustment handles and knobs shall not exceed 45° C. (ambient 23° C.).

3.4.3 *Base.* — The base shall be a metal casting of rugged design and sufficient weight to support the lamp housing properly. It shall permit a full range of adjustments and rigidly hold the lamp house at any desired angle without tipping.

3.4.4 *Extension cord.*—The electric extension cord shall be as specified in 3.3.5.1.

3.4.5 *Filters.*—A circular blue glass with 30 to 40 percent transmission and a circular ground (fine aluminum oxide abrasive) glass filter shall be supplied with the illuminator. Filters shall include a heat-absorbing glass permanently mounted in lens system.

3.4.6 *Iris diaphragm.*—The iris shall work freely and close to at least a 5-mm. diameter opening and shall be frictionally fitted to lock into position with a set screw. The iris shall be operated by a collar control

3.4.7 *Lamp.*—The illuminator shall accommodate, without adaptors, a G–16½, 115-volt, 110-watt incandescent bulb which shall be 3 inches + 0 − ⅛ inch in overall length, with single-contact bayonet candelabra base, light center of 1⅜ ± 1/32 inches, class C, CC–13 filament, with approximately 1600 lumens and shall have an average life of 200 hours.

3.4.8 *Illumination.* — Light shall be sufficiently bright for proper illumination of the microscope field when using a darkfield condenser. The lamp, when 10 inches from the microscope, shall give a filament image approximately 1¼ by 1¼ inches. Unless otherwise specified, the filament shall be the ribbon type. Illuminator shall be of such design that Critical and Kohler illumination can be used with the microscope. The temperature in the beam of light, with the

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:51Z
Check the source to verify that this is the current version before use.

illuminator focused at approximately 6 inches shall not exceed 70° C. (ambient 23° C.). The intensity of the light shall be not less than 300 foot candles measured at 6 inches from the front of the illuminator lens, with the blue and ground glasses in position, and with the illuminator focused at infinity.

3.4.9 *Lens system.*—Focusing of the lamp shall be accomplished by a positive rack and pinion with dual control buttons for operation from both sides. The two-lens system shall be designed with clear aperture of 60 mm. Range of focus shall be from 6 inches to infinity.

3.4.10 *Tilting.*—Angular tilting in the vertical plane shall be accomplished by a horizontal screw in the base with operating buttons cool at all times.

3.4.11 *Finish.* — Lamp housing and base shall be finished in black wrinkled enamel. All other parts subject to handling and wearing conditions shall be finished in accordance with best mechanical and commercial practice.

3.4.12 *Accessories.*—A demountable multiple-filter holder suitable for fitting to the front of the lamp shall be supplied. Filter holder shall be capable of holding four square filters, and shall be adjustable to take either 50-mm. square filters 4.0 to 5.0 mm. in thickness or 75-mm. square filters not exceeding 3.2 mm. in thickness. Unless otherwise specified, square filters shall not be furnished. An extra pair of round filters, blue and ground glass, and an extra G–16½ bulb shall be supplied (see 3 4.5 and 3.4.7).

3.4.13 *Carrying case.* — The illuminator with attached filter holder shall be packed in a hinged, hardwood, microscope-type case, approximately 12 inches long, 8½ inches wide, and 8½ inches high overall. Case shall be constructed of approximately ½-inch hardwood. Ends shall be grooved, glued, joined, and assembled by screws. Bottom screws shall be flathead and countersunk. Lid shall be flush type, approximately 1½ inches high, and shall be furnished with two stop-type, four-knuckle hinges, approximately 1 to 1½ inches long. A trunk-type lock latch, approximately 1¼ by 1¾ inches, shall be affixed to the front of the box by two screws on each latch member. Two keys shall be supplied for each latch. A suitable leather-covered handle approximately 5 inches long shall be affixed to the top of the center of the lid by means of two corrosion-resisting-metal handle loops. Inside of case shall be fitted at the bottom with formed wood members at each of the four corners so that the illuminator may be properly positioned and held in place. The inside top of the case shall be fitted at the back with two compartments, one at each end, to accommodate the glass filters. One compartment shall be suitable for holding four 2½-inch diameter round filters, and the other shall be so constructed that it shall hold three 2- by 2-inch or 3- by 3-inch square filters. Top of lid shall be fitted with a rubber-faced bumper to keep illuminator in position when lid is closed. Case shall be given a walnut finish.

3.5 Type IV.—Adjustable, darkfield illumination and general use, a.c.

3.5.1 *Material.*—Material shall be as hereinafter specified, and shall be of high quality throughout, suitable in all respects for the purpose intended.

3.5.2 *Design.* — Illuminator shall be used with a transformer (see 3.5.11), giving a concentrated light source suitable for opaque, bright, and darkfield illumination for use in routine and advanced medical and clinical laboratory procedures. The illuminator shall be compact, standing not over 3½ inches high, and 5½ inches long when focused at approximately 3 inches. The illuminator shall consist of a sturdy base; a three-link jackknife upright support; a three-element (lens) condenser in mount; a lamp housing; a filter holder with a ground glass and a blue filter; a lamp, socket with cord; and a trans-

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:51Z
Check the source to verify that this is the current version before use.

former with cord. The socket shall detach easily from the lamp housing for lamp replacement. The complete assembly shall easily detach from the base.

3.5.3 *Base* —Base shall be of cast iron with sufficient weight to give the illuminator stability in all positions of use. The base shall fall within a circle, 4½ inches in diameter.

3.5.4 *Jackknife upright support.* — The three-link jackknife upright support shall be of corrosion-resistant steel and shall be of such construction as to permit the illuminator to be adjusted from 8½ inches to 2¼ inches measured from the table to the center of the lens system.

3.5.5 *Lens system.* — The three-lens condensing system shall be of high quality suitable for the purpose intended The glass shall be free from any defects which might interfere with good optical imagery. The glass surfaces shall be ground and polished to the necessary accuracy for a high standard of performance The complete condensing system shall be housed in a suitable sleeve-type retainer and shall be focused by means of a spiral action in the lamp housing. The range of focus shall be from 3 inches (from the front element) to infinity. When focused at 3 inches, the illuminator shall give a uniform approximately 2-inch-diameter spot of light at approximately 6 inches from the front of the condenser mounting. The temperature in the beam with the illuminator focused at 6 inches with no filters shall be not more than 44° C. and under the same conditions but with the blue glass filter inserted, the temperature shall be not more than 39° C. (ambient 23° C.).

3.5.6 *Lamp housing.*—Lamp housing shall be constructed of aluminum, approximately 2¾ inches long and 1¾ inches in diameter. It shall be well ventilated and provided with cooling fins over the surface to dissipate the heat. Exterior parts of the illuminator shall not exceed 80° C. (ambient 23° C.).

3.5.7 *Filter holder.*—A cap-type aluminum filter holder shall be frictionally fitted on the condenser mounting.

3.5.8 *Filters.* — A circular ground-glass filter, and a circular clear-blue-glass filter with 20 to 25 percent transmission of white light shall be supplied.

3.5.9 *Lamp* —A 6.5 volt, 2.75 ampere, S–8 clear bulb, double-contact bayonet candelabra base, single filament type bulb shall be supplied. Overall length of the bulb shall be 2 inches.

3.5.10 *Socket, plugs, and cord.*—An electric extension cord approximately 3 feet long, leading from a permanent attachment in the lamp housing to the transformer, shall be provided. The cord shall be rubber insulated and shall have a two-pole polarity type male plug which fits into a corresponding female receptacle in the transformer. Another rubber cable wire, approximately 5½ feet long, shall lead from a permanent attachment in the transformer to a two-pole, parallel-blade male plug.

3.5.11 *Transformer.* — Overall dimensions of transformer shall be approximately 6 by 4½ by 3¾ inches. Primary voltage shall be 115 volts; secondary voltage shall be controllable between 4 to 6.8 volts. Volt-amperes shall be 30. Transformer shall operate on 60-cycle alternating current. Base of transformer shall have a suitable attachment for hanging. In addition, four rubber feet shall screw into the base.

3.5.12 *Finish.* — All metal parts shall be finished in baked black wrinkle enamel or chrome in accordance with the best commercial practice.

3.5.13 *Accessories.*—In addition to the bulb required as regular equipment, two extra bulbs shall be furnished as spare parts with each illuminator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

3.6 Operating instructions. — Detailed operating and maintenance instructions shall be provided with each illuminator.

3.7 Identification marking.—Illuminators shall be permanently and legibly marked to show the name or registered trade-mark of the manufacturer. Marking shall be accomplished by stamping, imprinting, or by use of a metal plate permanently affixed to the illuminator.

3.8 Workmanship. — Workmanship shall be first class throughout. Illuminators shall be free of defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, not less than 10 of each type illuminators shall be taken at random from each delivery.

4.2 Inspection.—Samples shall be inspected for material, design, measurements, performance, finish, inclusion of accessories, operating instructions, identification marking, workmanship, packaging, packing, and marking.

4.3 Tests.

4.3.1 *Color temperature correcting filters* — Spectrophotometer shall be used to determine transmission vs. wave length data. Color temperature altering capacity in mireds is evaluated from the slope of a plot of log T vs. λ.

$$\text{Mireds} = \Delta\left(\frac{1}{\theta}\right) = 160.7\left(\frac{\Delta \log T}{\Delta \frac{1}{\lambda}}\right)$$

θ = color-temperature ° Kelvin.
T = percent transmission.
λ = wave length (microns).

4.3.2 *Temperature in the beam of light.*— A thermocouple capable of measuring temperature with an accuracy of ± 2° C. shall be attached to the emulsion side of an exposed film having a density of 2.0. The emulsion side of the film shall be away from the light source. Measurements shall be made with the thermocouple 6 inches from the nearest lens with the lamp focused on the thermocouple. Aperture shall be at maximum. Lamps shall be continuously operated until a stable temperature is reached. Temperatures shall be actual readings not corrected for ambient temperature.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, each illuminator shall be packaged in accordance with accepted commercial practice.

5.2 Packing. — The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages* — Unless otherwise specified, each package shall be marked with the name and type of the commodity, stock number, quantity contained therein, and the names of the contractor and manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, each shipping container shall be marked with the name of the commodity as defined by the contract or order under which shipment is made, the quantity contained therein, name of the contractor, number of the contract or order, name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data. — Purchasers should specify the type, finish, and filters required, and should exercise any desired options offered herein. (See 1.2.1, 3.2.2.5, 3.3.8, 3.4.8, 3.4.12, 4.1, 5.1, 5.3, and 7.2.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

6.2 **Bid samples.**—It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification covers the following items listed in the Armed Services Catalog of Medical Materiel:

4–287–010—Illuminator, Microscope, Adjustable, 110 Volt, AC.
4–287–040—Illuminator, Microscope, Junior, 110 Volt, AC-DC.
4–287–050—Illuminator, Microscope, Senior, 110 Volt, AC-DC.
4–287–065—Illuminator, Microscope, Substage, Daylight, 110 Volt, AC-DC.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., normally procured to meet Federal Government requirements.

6.5 **Transportation description.**

6.5.1 *Transportation description applicable to these items is:*

Lamps, electric, not otherwise indexed by name, with globes or shades not exceeding 21 inches in diameter.

In barrels or in packages 5F, 30F, 219, 532, 817 or 970.
Carload weight 14,000 pounds, subject to Rule 34, Consolidated Freight Classification.
In barrels or boxes.
Truckload weight 14,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments:

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications.

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Specifications

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.
JAN–P–133—Packaging and Packing for Overseas Shipment — Boxes, Set-Up, Paperboard.
MIL–A–140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

[1] Applicable to Navy purchases only.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:51Z
Check the source to verify that this is the current version before use.

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

7.2.2 *Packaging.*—One illuminator of the type specified constitutes one unit. Each unit shall be packaged in a carton of appropriate size conforming to the requirements of Federal Specification LLL-B-631 or LLL-B-636, or Military Specification JAN-P-133. Each carton shall be adequately secured to prevent accidental opening. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents Care shall be exercised to insure that the bulbs are adequately protected against damage.

7.2.3 *Packing*

7.2.3.1 *Domestic* — The number of unit cartons specified in table I shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-621.

7.2.3.2 *Overseas* — The number of unit cartons specified in table I shall be packed in a shipping container designed for type 3 load and conforming to the requirements of Military Specification JAN-P-106. Closure and strapping shall conform to the requirements of the appendix thereto. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Military Specification JAN-P-125, for case liners; seams and closures shall be sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification MIL-A-140.

TABLE I.—*Packing quantities*

Stock number	Number of unit cartons to be packed in each shipping container
4-287-010----	6
4-287-040----	12
4-287-050----	1
4-287-065----	48

7.2.3.3 *Packing variation.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design and so constructed as to insure safe delivery to specified destination.

7.2.4 *Marking.*—Each unit carton and each shipping container shall be marked in accordance with Military Standard MIL-STD-129

7.2.5 *Inspection procedure (Navy only).*—The general inspection procedures for Naval purchases shall be in accordance with General Specifications for Inspection of Material.

7.2.6 *Ordering data.* — Procurement documents should specify whether the subject commodity is to be packed for domestic or overseas shipment (see 7.2.3).

NAVY INTEREST: Md

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 10 cents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.